

AI IELTS Instructor & Examiner

Complete System Architecture, AI Models, Data Pipeline & Training Strategy

1. Vision

Build a fully autonomous AI IELTS Instructor and Examiner capable of:

- Generating IELTS-style exams
 - Conducting speaking interviews
 - Evaluating writing tasks
 - Evaluating listening and reading performance
 - Identifying weaknesses
 - Providing personalized feedback
 - Acting as a 24/7 IELTS tutor
 - Continuously adapting to student skill level
-

2. Core Product Features

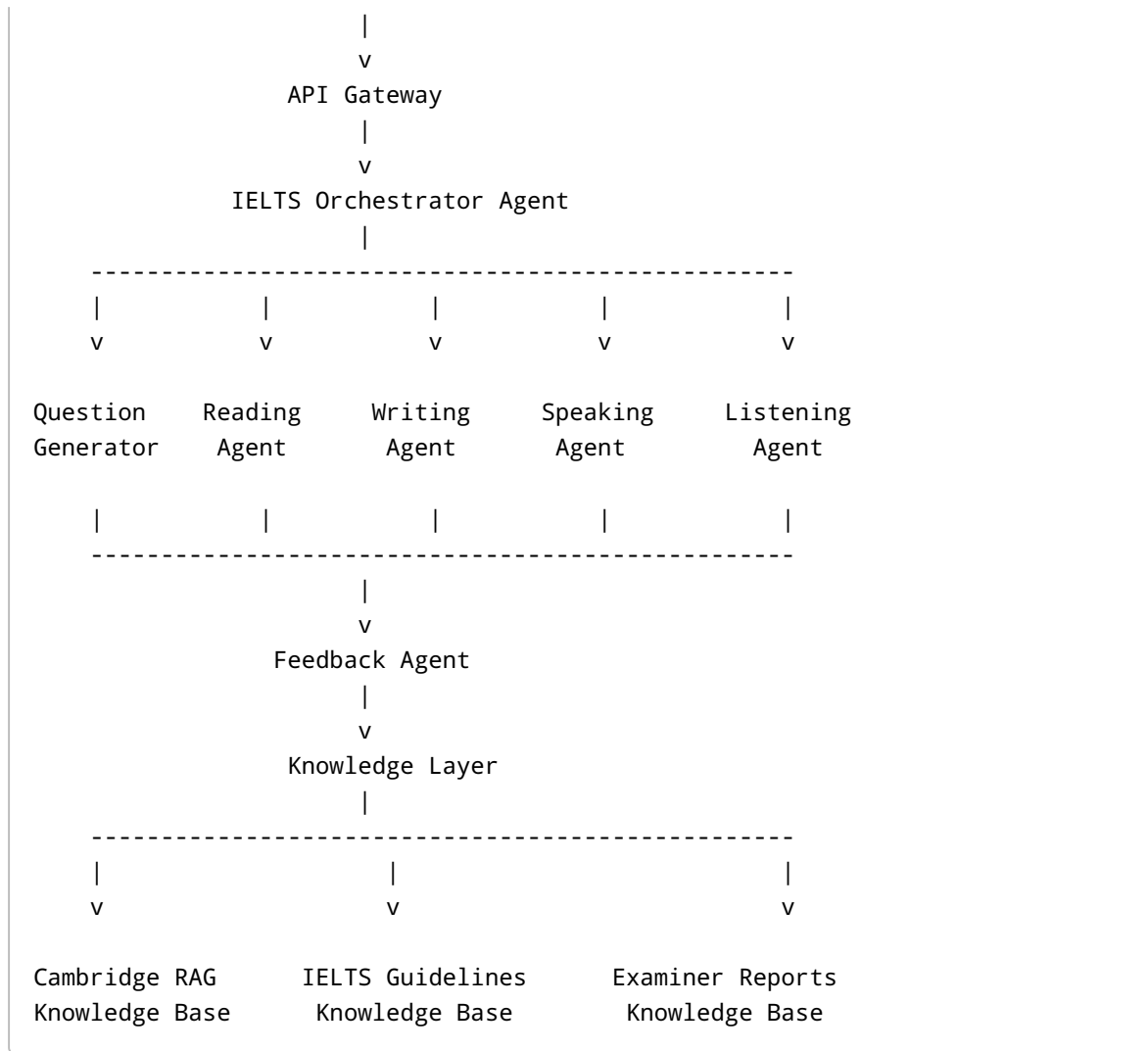
Student Dashboard

Features

- AI IELTS Instructor
 - AI Speaking Examiner
 - AI Writing Examiner
 - AI Reading Trainer
 - AI Listening Trainer
 - Progress Tracking
 - Mock Exams
 - Performance Analytics
 - Personalized Feedback
-

3. System Architecture

```
graph TD; Student --> FrontendLayer["Frontend Layer (Web + Mobile)"]
```



4. AI Agent Architecture

4.1 IELTS Orchestrator Agent

Responsibilities:

- Controls all workflows
- Routes tasks
- Coordinates agents
- Combines outputs

Frameworks:

- LangGraph
- CrewAI
- AutoGen

4.2 Question Generator Agent

Generates:

Reading

- Matching Headings
- TFNG
- MCQ
- Sentence Completion

Listening

- Maps
- Forms
- Notes
- Tables

Writing

- Task 1
- Task 2

Speaking

- Part 1
- Part 2
- Part 3

Output:

```
{
  "question_type": "Writing Task 2",
  "difficulty": "Band 7",
  "question": "Some people believe..."
}
```

4.3 Writing Examiner Agent

Responsibilities:

- Essay evaluation
- Band prediction
- Grammar analysis
- Vocabulary analysis
- Coherence analysis

Outputs:

```
{
  "band_score": 6.5,
  "grammar": 6,
  "vocabulary": 7,
  "coherence": 6,
  "task_response": 7
}
```

4.4 Speaking Examiner Agent

Responsibilities:

- Conduct interview
- Evaluate pronunciation
- Evaluate fluency
- Evaluate grammar
- Predict IELTS band

Outputs:

```
{
  "band_score": 7,
  "pronunciation": 7,
  "fluency": 7,
  "grammar": 6.5,
  "vocabulary": 7
}
```

4.5 Feedback Agent

Responsibilities:

- Explain mistakes
- Suggest improvements
- Generate study tasks
- Provide coaching

Example:

You repeatedly used basic vocabulary.

Replace:
good

With:
beneficial
effective
valuable
advantageous

5. Knowledge Layer

Purpose

Provide factual IELTS knowledge to AI.

NOT used for direct training.

Used through Retrieval-Augmented Generation (RAG).

6. Knowledge Sources

Cambridge IELTS

- Cambridge IELTS 1–19
-

Official IELTS Resources

- Official IELTS Guide
 - British Council Resources
 - IDP Resources
-

Examiner Reports

Contains:

- Band descriptors
 - Common mistakes
 - Scoring criteria
-

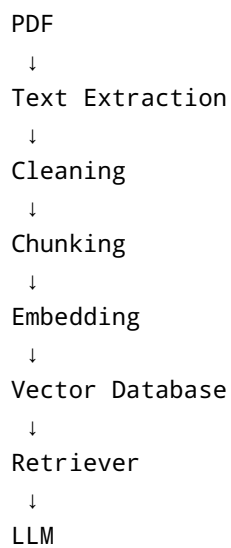
Vocabulary Resources

Contains:

- Academic vocabulary
 - Band 7-9 vocabulary
 - Synonym databases
-

7. RAG Architecture

Workflow



```
graph TD; A[PDF] --> B[Text Extraction]; B --> C[Cleaning]; C --> D[Chunking]; D --> E[Embedding]; E --> F[Vector Database]; F --> G[Retriever]; G --> H[LLM];
```

PDF
↓
Text Extraction
↓
Cleaning
↓
Chunking
↓
Embedding
↓
Vector Database
↓
Retriever
↓
LLM

Why RAG?

Benefits:

- Easy updates
 - Lower costs
 - Higher accuracy
 - No retraining required
-

8. Document Processing Pipeline

Extraction

Tools:

- PyMuPDF
 - PDFPlumber
-

Cleaning

Remove:

- Headers
 - Footers
 - Page numbers
 - Formatting artifacts
-

Chunking

Chunk Size:

500–800 Tokens

Overlap:

100 Tokens

9. Embedding Models

Recommended

BGE-M3

Best overall choice

Alternative

- BGE Large
- E5 Large

- OpenAI Embeddings
-

10. Vector Database

Preferred

Qdrant

Advantages:

- Open source
- Fast retrieval
- Cheap deployment

Alternatives:

- Pinecone
 - Weaviate
-

11. Main Instructor Model

Base Model

Qwen 3 32B

Reason:

- Strong instruction following
 - Excellent reasoning
 - Open source
-

Future Upgrade

Qwen 3 72B

or

Llama 3.3 70B

12. Fine-Tuning Strategy

Goal

Create an IELTS-specialized instructor.

Train On

Task 1

Question Generation

Examples:

```
{
  "instruction": "Generate IELTS Speaking Part 2 Question",
  "output": "Describe a memorable journey..."
}
```

Task 2

Answer Evaluation

Examples:

```
{
  "essay": "...",
  "band_score": 7,
  "feedback": "Good vocabulary..."
}
```

Task 3

Error Explanation

Examples:

```
{
  "question": "...",
  "student_answer": "B",
  "correct_answer": "C",
}
```

```
"explanation": "Option C is correct because..."
}
```

13. Writing Evaluation System

Model

DeBERTa-v3 Large

Fine-Tuned

Inputs

Essay
Prompt

Outputs

Band Score
Task Response
Grammar
Vocabulary
Coherence

14. Speaking Evaluation System

Pipeline

```
Audio
↓
Whisper
↓
Transcript
↓
Feature Extraction
↓
Scoring Model
```

Models

Speech Recognition

Whisper

Audio Understanding

Wav2Vec2

Band Prediction

XGBoost

15. Weakness Detection System

Predict:

- Grammar Problems
- Vocabulary Problems
- Coherence Problems
- Pronunciation Problems
- Fluency Problems

Outputs:

```
{
  "grammar": true,
  "vocabulary": false,
  "coherence": true,
  "pronunciation": false
}
```

16. Data Sources

Cambridge IELTS

Used For:

- Question patterns
 - Difficulty calibration
 - Exam structure
-

Public Datasets

Writing

- ASAP Essay Dataset
- Kaggle Essay Scoring

Speech

- Common Voice
 - TED-LIUM
 - Speech Accent Archive
-

Synthetic Data

Generated by AI

Examples:

- Essays
- Speaking responses
- Reading questions
- Listening questions

Band Range:

Band 1–9

17. Training Pipeline

```
Raw Data
  ↓
Cleaning
  ↓
Labeling
  ↓
Instruction Dataset Creation
  ↓
Fine-Tuning
  ↓
Evaluation
  ↓
Deployment
```

18. Evaluation Metrics

Writing

- MAE
- RMSE
- QWK

Target:

QWK > 0.85

Speaking

- Band Accuracy
 - MAE
 - Pearson Correlation
-

RAG

- Recall@K
 - MRR
 - Faithfulness
 - Answer Relevance
-

19. Backend Stack

API

FastAPI

Authentication

JWT

OAuth

Database

PostgreSQL

Vector Database

Qdrant

Storage

S3

20. Frontend Stack

Web

Next.js

React

Mobile

Flutter

21. Deployment

Containers

Docker

Orchestration

Kubernetes

Monitoring

Prometheus

Grafana

22. Future Roadmap

Phase 1

- RAG Knowledge Base
- AI Instructor Chat

Phase 2

- Writing Evaluation

Phase 3

- Speaking Examiner

Phase 4

- Full Mock Test Generator

Phase 5

- Adaptive Learning System

Phase 6

- Multi-Agent IELTS Ecosystem

Final Architecture

```
Qwen 3 Fine-Tuned Instructor
+
Cambridge RAG Knowledge Base
+
Writing Evaluation Models
+
Speaking Evaluation Models
+
Feedback Engine
+
Multi-Agent Orchestration
+
Adaptive Exam Generator
=
AI IELTS Instructor & Examiner
```